

Question No:5

DS	Session	Tree 1	Question Information	Level 1	Challenge 65
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Question description

Given an array of n integers, your task is to process q queries of the form: what is the minimum value in range [a,b]?

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Input

The first input line has two integers n and q: the number of values and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b: what is the minimum value in range [a,b]?

Output

Print the result of each query.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3 2 4 5 1 1 5 3
2 4
5 6
1 8
3 3
```

EXPECTED OUTPUT

```
2
1
1
4
```

Test Case 2

INPUT (STDIN)

```
13 12 14 51 11 1 5 3 4 6
3 4
3 6
4 8
1 3
```

EXPECTED OUTPUT

```
14
1
1
12
```

Code:

```
1  #include <stdio.h>
2  #define MAX 100005
3  #define INF 1000000000
4
5  int tree[4 * MAX];
6  int a[MAX];
7
8  void build(int *aa, int k, int l, int r) {
9      if (l == r) {
10         tree[k] = aa[l];
11         return;
12     }
13
14     int mid = (l + r) / 2;
15
16     build(aa, 2 * k, l, mid);
17     build(aa, 2 * k + 1, mid + 1, r);
18
19     if (tree[2 * k] < tree[2 * k + 1])
20         tree[k] = tree[2 * k];
21     else
22         tree[k] = tree[2 * k + 1];
23 }
24
25 int query(int k, int l, int r, int ql, int qr) {
26     if (qr < l || r < ql)
27         return INF;
```

```

29     if (ql <= l && r <= qr)
30         return tree[k];
31
32     int mid = (l + r) / 2;
33
34     int left = query(2 * k, l, mid, ql, qr);
35     int right = query(2 * k + 1, mid + 1, r, ql, qr);
36
37     return (left < right) ? left : right;
38 }
39
40 int main() {
41     int n, q;
42
43     scanf("%d %d", &n, &q);
44
45     for (int i = 1; i <= n; i++)
46         scanf("%d", &a[i]);
47
48     build(a, 1, 1, n);
49
50     while (q--) {
51         int l, r;
52
53         scanf("%d %d", &l, &r);
54
55         printf("%d\n", query(1, 1, n, l, r));
56     }
57
58     return 0;
59 }

```

Output:
Test Case-1:

	8 4	
	3 2 4 5 1 1 5 3	
	2 4	
	2	
	5 6	
	1	
	1 8	
	1	
	3 3	
	4	
	8 6 4 5 1 1 5 3	

Test Case-2:

	10 4	
	13 12 14 51 11 1 5 3 4 6	
	3 4	
	14	
	3 6	
	1	
	4 8	
	1	
	1 3	
	12	

Keshav Kumar Rauniyar
CH.SC.U4CSE25236